

TAICHUNG SHUINAN AP, Taiwan

WMO: 467510

Lat: 24.186N

Lon: 120.654E

Elev: 115

StdP: 99.95

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	9.2	4.9	5.4	9.3	6.2	5.9	10.4	10.1	16.3	9.2	15.9	1.8	10	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	34.2	27.8	33.9	27.8	33.1	27.5	29.0	33.2	28.4	32.7	27.9	32.2	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	24.3	32.4	27.2	23.2	31.7	26.8	22.8	31.4	95.5	33.1	92.5	32.8	90.4	32.3	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.0	7.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			3.5	30.1	2.3	0.5	1.8	30.5	0.5	30.8	-0.8	31.1	-2.4	31.4
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	16.9	17.3	19.4	23.3	26.0	28.1	29.1	29.0	27.6	25.2	21.9	18.4	(6)
(7)		DBStd	5.04	2.58	3.07	3.53	3.23	2.23	1.75	1.22	1.37	1.73	1.95	2.59	2.54	(7)
(8)		HDD10.0	1	0	0	1	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	181	58	52	31	5	0	0	0	0	0	0	4	31	(9)
(10)		CDD10.0	4941	215	204	292	397	495	542	593	589	527	470	358	259	(10)
(11)		CDD18.3	2080	14	22	64	153	237	292	335	331	277	212	112	32	(11)
(12)		CDH23.3	19708	81	129	354	1067	2024	3171	3972	3858	2763	1540	587	162	(12)
(13)		CDH26.7	7628	7	19	82	315	725	1303	1751	1657	1084	527	144	14	(13)
(14)	Wind	WSAvg	2.9	3.6	3.3	3.2	2.7	2.3	2.6	2.3	2.5	2.7	3.4	3.5	3.4	(14)
(15)	Precipitation	PrecAvg	1603	33	77	95	116	214	347	233	315	108	15	17	21	(15)
(16)		PrecMax	2338	123	384	352	473	541	890	643	785	576	129	74	84	(16)
(17)		PrecMin	979	4	3	3	0	6	46	21	74	5	0	0	0	(17)
(18)		PrecStd	353	27	94	72	99	120	218	151	170	108	23	21	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.6	28.8	30.3	32.2	33.9	35.0	35.2	35.2	34.2	33.0	31.2	28.2	(19)
(20)		MCWB	21.3	22.6	23.8	26.1	27.2	27.8	28.2	28.3	27.9	26.6	25.3	22.6	(20)	
(21)		2%	DB	25.6	26.9	28.9	31.1	32.4	33.9	34.2	34.2	33.2	31.8	29.9	26.8	(21)
(22)		MCWB	20.0	21.0	23.4	25.6	26.6	27.7	27.9	27.9	27.7	26.1	24.1	21.1	(22)	
(23)		5%	DB	23.9	24.9	27.2	29.9	31.5	33.1	33.7	33.4	32.6	30.9	28.2	25.1	(23)
(24)		MCWB	18.9	20.1	22.4	24.9	26.3	27.3	27.8	27.7	27.3	25.5	23.2	20.1	(24)	
(25)		10%	DB	22.0	22.9	25.2	28.7	30.6	32.1	33.0	32.8	31.8	29.8	26.9	23.9	(25)
(26)		MCWB	17.9	19.1	21.3	24.3	26.0	27.0	27.6	27.5	26.8	24.7	22.2	19.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.4	23.6	25.0	27.1	28.2	29.2	29.7	29.6	29.0	27.6	26.3	23.0	(27)
(28)		MDCB	26.8	27.7	29.4	31.1	32.2	33.4	33.6	33.7	32.8	31.6	30.2	27.6	(28)	
(29)		2%	WB	20.4	21.9	23.8	26.1	27.2	28.3	29.0	28.7	28.2	26.6	24.6	21.7	(29)
(30)		MDCB	24.4	25.4	27.9	30.2	31.3	32.7	33.2	33.2	32.3	30.6	28.8	25.7	(30)	
(31)		5%	WB	19.3	20.6	22.8	25.2	26.6	27.8	28.4	28.2	27.7	25.9	23.6	20.7	(31)
(32)		MDCB	23.1	24.1	26.6	29.1	30.5	32.1	32.7	32.6	31.8	29.8	27.5	24.5	(32)	
(33)		10%	WB	18.3	19.3	21.7	24.4	26.0	27.2	27.7	27.7	27.0	25.2	22.6	19.6	(33)
(34)		MDCB	21.7	22.6	24.7	27.9	29.7	31.3	32.0	31.9	30.9	29.0	26.1	22.9	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.3	10.1	9.3	8.7	8.2	7.8	8.0	7.7	8.4	9.5	9.9	10.8	(35)	
(36)		5% DB	MDCDBR	12.8	13.1	11.9	10.0	9.3	8.8	8.7	9.0	10.4	11.1	12.5	(36)	
(37)		MCWBR	8.2	8.0	7.3	5.8	5.1	4.7	4.6	4.3	5.0	6.0	6.7	7.9	(37)	
(38)		5% WB	MDCDBR	11.8	12.1	11.5	9.5	8.8	8.4	8.5	8.3	8.5	9.3	10.7	11.6	(38)
(39)	MCWBR	8.0	7.9	7.4	6.0	5.3	4.9	4.8	4.6	5.0	5.8	6.7	7.9	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.491	0.546	0.666	0.677	0.580	0.533	0.546	0.580	0.572	0.535	0.501	0.492	(40)	
(41)		taud	2.009	1.893	1.652	1.682	1.945	2.094	2.047	1.943	1.939	1.986	2.065	2.024	(41)	
(42)		Ebn at Noon	752	740	673	679	746	777	766	741	736	740	739	732	(42)	
(43)		Edh at Noon	158	189	252	249	191	163	171	189	186	170	148	150	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.90	3.21	3.69	4.06	4.65	5.13	5.84	5.29	4.87	4.30	3.23	2.87	(44)	
(45)		RadStd	0.40	0.47	0.41	0.55	0.70	0.56	0.36	0.51	0.46	0.36	0.31	0.35	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.25	N/A	N/A	+0.22	N/A	N/A	N/A	N/A	+127	+112

Nomenclature: See separate page